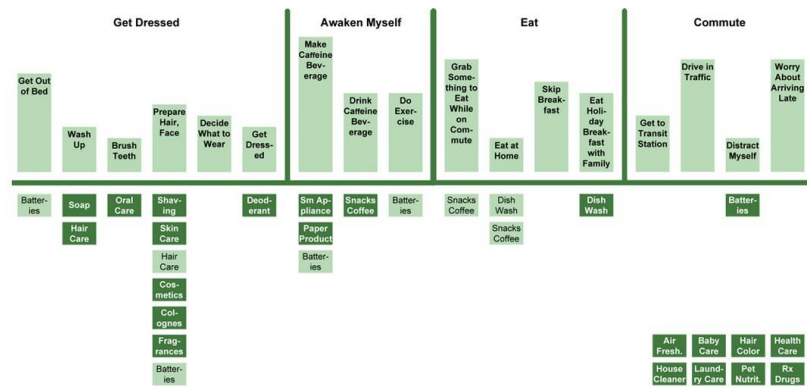




https://www.interaction-design.org/literature/article/a-very-useful-work-of-fiction-mental-models-in-design?utm_source=twitter&utm_medium=sm

A Very Useful Work of Fiction – Mental Models in Design

Mental models play an important role in Human-Computer Interaction (HCI) and **interaction design**. They relate to the way that a user perceives the world around them and are based in belief as opposed to being a factual concept. However, if you can understand your users' **mental models**, you can simulate these models within your designs to make them more usable and intuitive.

Mental models are an artefact of belief. They are the beliefs that a user holds about any given system or interaction. In most instances, the belief will – to a certain extent – resemble the real life model. This is important because users will plan and predict future actions within a system based on their mental models.

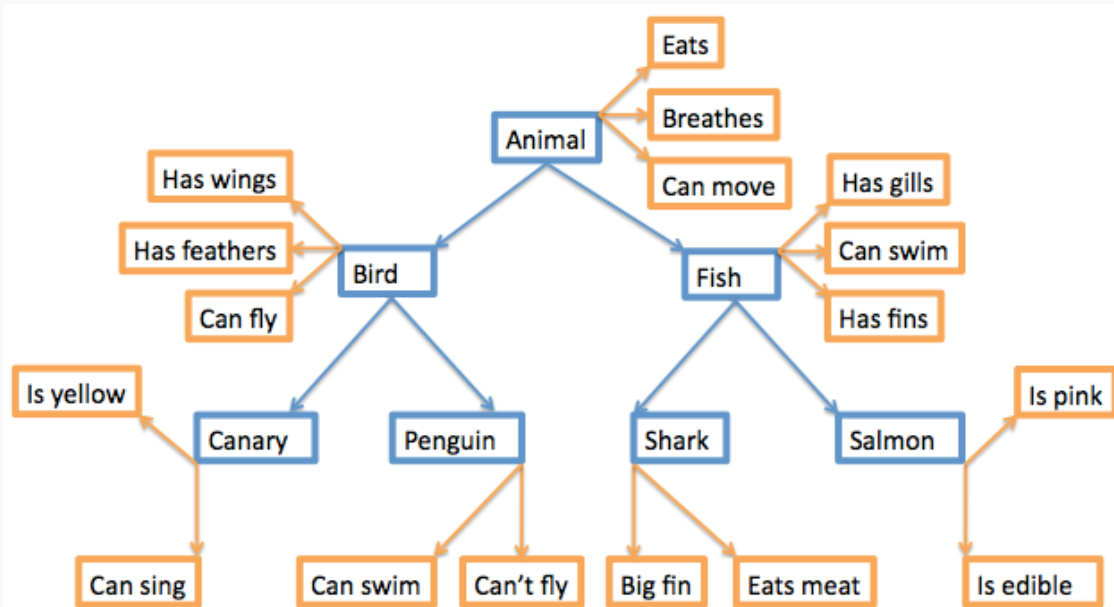
Designers can tap into users mental models so that their products communicate their function through their form. However, they can only do this successfully if they truly understand their users' mental models. It is an all too common failing of designs for designers to base their ideas on their own mental models; their models are often too complete and detailed to bear any relationship with a user's model. This in turn leads to failure in UI where the user does not find their mental model and is left confused and frustrated.

It is worth noting that a mental model is not a static creation. It is capable of evolution and may change based on using other products, from interacting with other users or taking on board learning from other sources.

The concept of mental models comes from the Scottish psychologist, Kenneth Craik's, book "The Nature of Explanation." He said that the mind constructs "small-scale models of reality" to anticipate and explain events.

The Mental Model Rule of Thumb

Given that we all have mental models of interaction – it is a good rule of thumb to assume that wherever possible; users will form their mental models based on interactions with existing applications and web sites. In short, they expect functionality to be consistent with these previous experiences and wherever a UI standard pattern exists, it should be emulated in your designs.

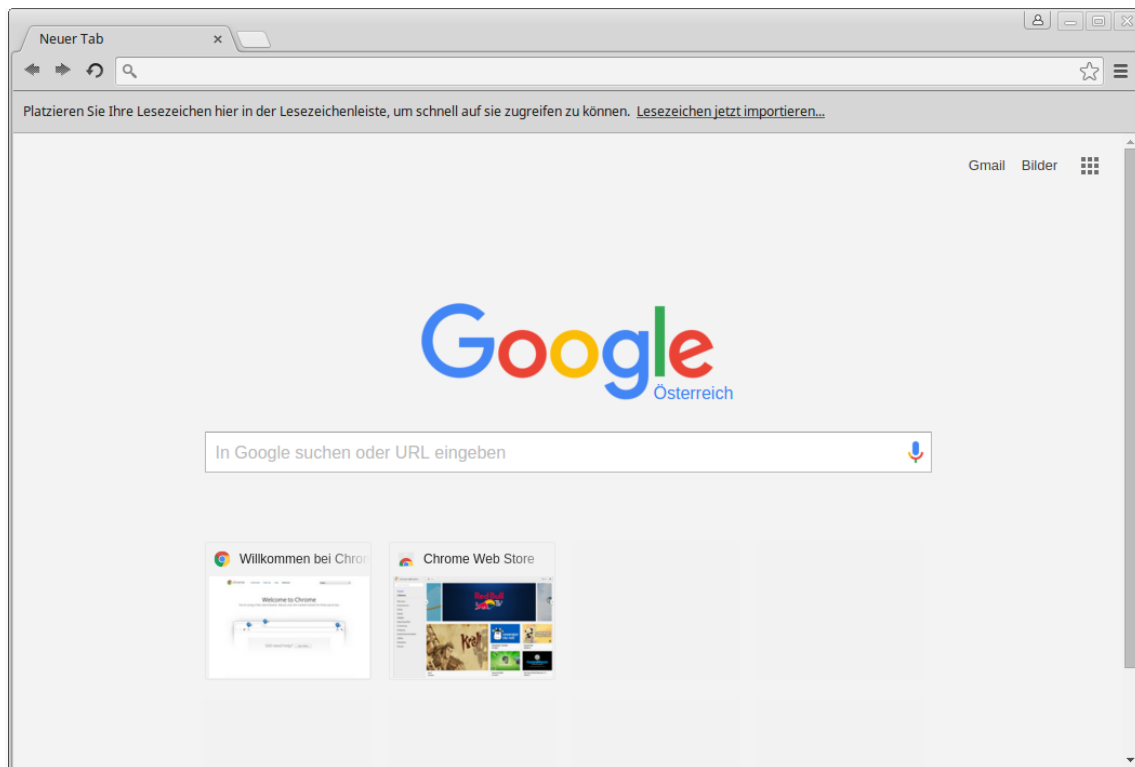


Author/Copyright holder: Nathanael Crawford. Copyright terms and licence: CC BY-SA 3.0

Confused Mental Models

Jakob Nielsen, the founding member of the Nielsen Norman Group, offers an excellent insight into potential confusion in mental modelling. This takes place when users fail to recognise key differences between similar (but not identical) parts of a system.

He observes that many users seem unable to distinguish between the search field on Google and the URL entry field in a browser. These users will use Google to search for the name of a well-known website in order to click the link rather than simply entering the name plus “.com” in the URL bar. (A problem that Google Chrome, shown below, tries to resolve by making the URL bar a search tool).



Author/Copyright holder: Axendo. Copyright terms and licence: Public Domain.

It appears that users know that they can type stuff into a box and get taken to where they want to go but are uncertain as to the functionality of any given box. This offers an explanation as to why browsers have evolved to treat the URL box as a **search box** in many cases now. The functionality of the browser had to change to meet the user's mental model as the mental model itself was too confused to easily adopt the design pattern that users encountered.

Neilsen also offers a series of common mental model confusions that some users appear to encounter on a regular basis. They do not understand the differences between:

- Windows provided inside an operating system and those inside a browser
- Windows and applications
- Icons and applications
- Browser commands and commands that are held within an app
- Local and remote information
- Log in options and passwords (e.g. a user will often log into any website as though they were trying to log in to their e-mail account)

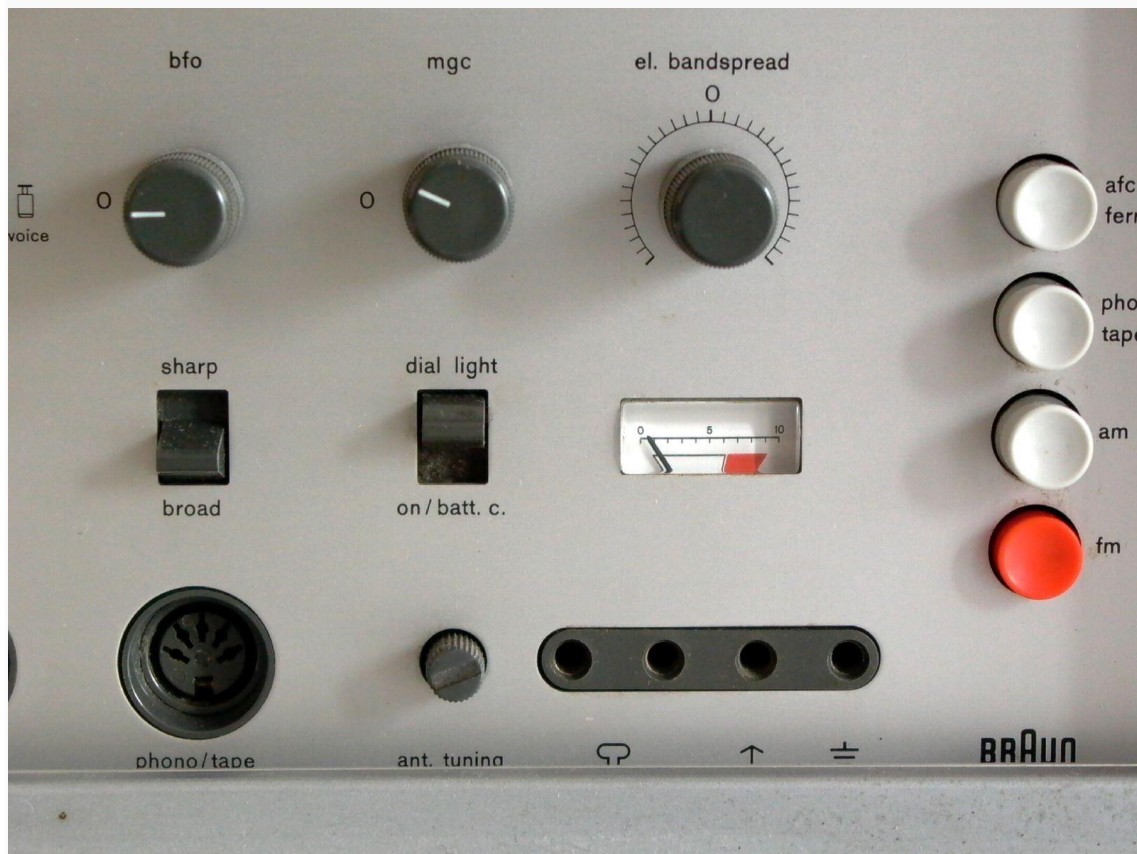
Taking Advantage of Mental Models in Design

It's worth noting that there is a level of inertia that must be overcome if you want to try to change someone's mental model. They have developed an understanding (whether this understanding is flawed or not) of a concept – replacing this takes time and effort.

If you are going to challenge a mental model and try to supplant it, you need to be aware of the intense efforts required to overcome this inertia. The user must perceive the value they get from your product as of “high value” to themselves – otherwise it is likely that they may become frustrated and go elsewhere where the environment is tailored to match their mental model.

The easiest course of action, when you are confronted with a clash between the design and a user’s mental model, is to alter the design to match the user’s model. In the vast majority of cases this is going to be the best action to take; you will not have to confront that inertia and the user will thank you for it.

If it is not possible to change the design to meet the model; you should take additional steps to help the user’s mental model come to terms with the new model. That means providing instructions, labels, tutorials and **visual cues** to help the new model become embedded in the user’s mind.



Author/Copyright holder: Nite_Owl. Copyright terms and licence: CC BY-SA 2.0

The Take Away

Mental models provide the designer with a shortcut when developing UI. By learning the user’s mental models and not relying on their own internal mental model a designer can emulate these models through the UI and create an experience which is intuitive and easy to follow. This in turn will result in a superior user-experience in which the user can focus on their task rather than

complex learning of new models. Mental models can be challenged but careful thought must be put into making the transition to a new model as easy as possible for the user.

References

A further explanation of mental models can be found here
- <https://www.interaction-design.org/literature/book...>

Jakob Nielsen's paper on Mental Models for the Nielsen Norman group can be found here - <https://www.nngroup.com/articles/mental-models/>

Hero Image: Author/Copyright holder: Rosenfeld Media. Copyright terms and licence: CC BY 2.0